Inverse of multivector: Beyond p+q=5 threshold

A. Acus* and A. Dargys**

*Institute of Theoretical Physics and Astronomy, Vilnius University,
Saulėtekio 3, LT-10257 Vilnius, Lithuania
**Center for Physical Sciences and Technology, Semiconductor Physics Institute,
Saulėtekio 3, LT-10257 Vilnius, Lithuania

August 7-11, 2017, Ghent

11$^{\mathrm{th}}$ International Conference on Clifford Algebras and their Applications in Mathematical Physics

Problem formulation and notations

- General Clifford algebra multivector (MV) $\mathsf{A} = \mathrm{scalar} + \mathrm{bivector} + \cdots + \mathrm{pseudoscalar}$ will be considered in this report
- The problem is to find inverse of general MV A, i. e. A^{-1} for which $\mathsf{A}\mathsf{A}^{-1} = \mathsf{A}^{-1}\mathsf{A} = 1$ holds,
- $n = p + q$ in the title indicates that formula for A^{-1} depends only on dimension of the Clifford algebra $Cl_{p,q}$ and not on particular signature (p, q)
- The search of solution of the problem heavily relies on computer-aided symbolic mathematics

So before explaining calculation of inverse of MV a brief outlook of the *Mathematica* package that was written for this purpose will be presented.

GeometricAlgebra package for *Mathematica*

Main features

- Textbook notation (implemented without *Mathematica* palettes) and automatic precedence of inner, outer and geometric products
- Algebraic operations in orthonormal frame (additive representation, symbolic coefficients)
- Switching between multiple algebras (denoted by different color) in same *Mathematica* session
- Matrix representations of $Cl_{p,q}$ algebras
- Idempotents with different base element sorting
- Main involutions and general multivector inverse

Files

- GAC_Base.nb (main file with installation information)
- UnicodeCharactersAdd.tr (add-on for *Mathematica* system file)
- Start.nb (first usage example with short introduction)
- AlgebraFormulary.nb (Clifford algebra formulary)
- ElementaryFunctions.nb (Clifford function expansions)
- Idempotents.nb (algebras idempotents)
- Blades (general blade formulas test)
- MultivectorInverse (general multivector inverses and formula test)
- MatrixRepresentations.nb (Clifford algebra matrix representations)

May be download under licence GPL-3.0 from `https://github.com/ArturasAcus/GeometricAlgebra`

Grade-negated self-product

Grade-r negation operation

- Changes the sign of grade-r part of the multivector A
- It is an involution denoted as $A_{\bar{r}}$. Note the bar over negated grade index.
- Formal definition $A_{\bar{r}} = A - 2\langle A\rangle_r$
- Main properties: $(A_{\bar{p}})_{\bar{r}} = A_{\bar{p},\bar{r}} = A_{\bar{r},\bar{p}}$, $A_{\bar{r},\bar{r}} = A$, $(A + B)_{\bar{r}} = A_{\bar{r}} + B_{\bar{r}}$
- However, $(AB)_{\bar{r}} \neq A_{\bar{r}}B_{\bar{r}}$
- Commutator $AA_{\bar{r}} - A_{\bar{r}}A = 2(\langle A\rangle_r A - A\langle A\rangle_r)$

Relation to standard involutions

- Reversion $\widetilde{A} = A_{\bar{2},\bar{3},\bar{6},\bar{7},\bar{10},\bar{11}\ldots}$
- Grade inversion $\widehat{A} = A_{\bar{1},\bar{3},\bar{5},\bar{7},\bar{9},\bar{11}\ldots}$
- Clifford conjugate $\widehat{\widetilde{A}} = A_{\bar{1},\bar{2},\bar{5},\bar{6},\bar{9},\bar{10}\ldots}$

Important: grades $0; 4; 8; 12; \cdots$ are absent in the above list of standard GA involutions. They play special role in inverse formulas for $n > 5$.

Grade-negated self-product

Definitions and the idea

Grade-negated self-product is defined via grade-r negation operation.

- Grade-negated self-product of general multivector A is geometric product of A with any number of its grade-negated counterparts. Examples $\mathsf{A}\mathsf{A}_{\bar{r}}\mathsf{A}_{\bar{k},\bar{l},\bar{t}}\mathsf{A}_{\bar{s},\bar{t}}\cdots$ or $\mathsf{A}_{\bar{r}}\mathsf{A}_{\bar{k},\bar{l},\bar{t}}\mathsf{A}_{\bar{s},\bar{t}}\cdots\mathsf{A}$. operation.
- Self-products where the initial MV A stands in the left-most or in the right-most position are of special importance in finding the inverse multivector A^{-1}.
- Any identity, which contains only operations of geometric product and grade negations, and which was derived in particular Clifford algebra of dimension $n = p + q$ will also be true in any n-dimensional algebra of arbitrary signature (p, q).
- Assume we can construct a scalar s_m using just grade-negated self-products of m multivectors with the initial multivector A standing either in the left-most or right-most position.

Grade-negated self-product

Definitions and the idea (continued)

- Name this way obtained scalar the discriminant of A.

$$\text{scalar}: \quad s_m = \overbrace{\mathrm{A}\mathrm{A}_{\bar{r},\bar{t},\ldots}\mathrm{A}_{\bar{s},\ldots}\cdots}^{m}$$

- If we succeed in constructing such a scalar then left/right inverse of MV can be obtained by removing left or right most positioned initial MV A from the scalar $s_m = \mathrm{A}\mathrm{A}_{\bar{r},\bar{t},\ldots}\mathrm{A}_{\bar{s},\ldots}\cdots$ and then dividing the result by s_m:

$$\mathrm{A}^{-1} = \frac{\mathrm{A}_{\bar{r},\bar{t},\ldots}\mathrm{A}_{\bar{s},\ldots}\cdots}{s_m}.$$

1. Introduction
2. Grade-negated self-product
3. Inverse of general MV of algebras of dimension $n = p + q \leq 5$
4. Inverse of general MV of algebras of dimension $n = p + q = 6$
5. Conclusions and perspective

Inverses in dimensions $n = p + q \leq 5$

Two dimensional algebras: $Cl_{2,0}$ case

- Write $\mathrm{A} = \sum_{J=0}^{2^n-1} a_J \mathbf{e}_J$, where the multi-index J covers all orthonormal base elements arranged in inverse degree lexicographic order, i. e., the lowest grade elements appear first while elements of the same grade are ordered lexicographically: $J = [\{\}, \{1\}, \{2\}, \{1, 2\}]$.
- Check that self-product $\mathrm{A}\mathrm{A}_{\bar{1},\bar{2}}$ immediately gives the required scalar $s_2 = a_{\{\}}^2 - a_1^2 - a_2^2 + a_{1,2}^2$.

Coefficient notation

- It is more convenient to rewrite the coefficient indices as $s_2 = a_0^2 - a_1^2 - a_2^2 + a_3^2$, i. e. the index of scalar a_0 is zero, the coefficients at vectors run from 1 to n while numerical indices of higher grades increase monotonically up to 2^n.
- General rule for grade-r index: start the element index by $\sum_{k=0}^{r-1} \binom{n}{k}$ and end by $\left(\sum_{k=0}^{r-1} \binom{n}{k}\right) + \binom{n}{r} - 1$. Here $\binom{n}{k}$ is the binomial coefficient.

Inverses in dimensions $n = p + q \leq 5$

Notes about discriminant

- The discriminant s_m comprises all coefficients a_i of multivector A
- The MV has inverse if discriminant is unequal to zero, $s_m \neq 0$. This imposes the requirement for MV coefficients.
- Because reverse leaves the discriminant invariant, the formula for $n = 2$ can be rewritten as $s_2 = \mathrm{A}\mathrm{A}_{\bar{1},\bar{2}} = \mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}$, from which the left/right inverse formulas follow,

$$\mathrm{A}^{-1} = \mathrm{A}_{\bar{1},\bar{2}}/(\mathrm{A}\mathrm{A}_{\bar{1},\bar{2}}); \qquad \mathrm{A}^{-1} = \mathrm{A}_{\bar{1},\bar{2}}/(\mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}).$$

- The eliminated grades are grade-negated in one of MVs in the self-product. In the example the grades 1 and 2 have been eliminated simultaneously by $\mathrm{A}_{\bar{1},\bar{2}}$. This rule is valid for discriminants up to $n \leq 5$.
- In all cases the highest degree of coefficients in the discriminant always matches the degree of determinant of real matrix representation of A, in this case 2×2 real matrices (next slide)

Inverses in dimensions $n = p + q \leq 5$

Notes about discriminant

- The discriminant s_m comprises all coefficients a_i of multivector A
- The MV has inverse if discriminant is unequal to zero, $s_m \neq 0$. This imposes the requirement for MV coefficients.
- Because reverse leaves the discriminant invariant, the formula for $n = 2$ can be rewritten as $s_2 = \mathrm{A}\mathrm{A}_{\bar{1},\bar{2}} = \mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}$, from which the left/right inverse formulas follow,

$$\mathrm{A}^{-1} = \mathrm{A}_{\bar{1},\bar{2}}/(\mathrm{A}\mathrm{A}_{\bar{1},\bar{2}}); \qquad \mathrm{A}^{-1} = \mathrm{A}_{\bar{1},\bar{2}}/(\mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}).$$

- The eliminated grades are grade-negated in one of MVs in the self-product. In the example the grades 1 and 2 have been eliminated simultaneously by $\mathrm{A}_{\bar{1},\bar{2}}$. This rule is valid for discriminants up to $n \leq 5$.
- In all cases the highest degree of coefficients in the discriminant always matches the degree of determinant of real matrix representation of A, in this case 2×2 real matrices (next slide)

Inverses in dimensions $n = p + q \leq 5$

Notes about discriminant

- The discriminant s_m comprises all coefficients a_i of multivector A
- The MV has inverse if discriminant is unequal to zero, $s_m \neq 0$. This imposes the requirement for MV coefficients.
- Because reverse leaves the discriminant invariant, the formula for $n = 2$ can be rewritten as $s_2 = \mathrm{A}\mathrm{A}_{\bar{1},\bar{2}} = \mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}$, from which the left/right inverse formulas follow,

$$\mathrm{A}^{-1} = \mathrm{A}_{\bar{1},\bar{2}}/(\mathrm{A}\mathrm{A}_{\bar{1},\bar{2}}); \qquad \mathrm{A}^{-1} = \mathrm{A}_{\bar{1},\bar{2}}/(\mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}).$$

- The eliminated grades are grade-negated in one of MVs in the self-product. In the example the grades 1 and 2 have been eliminated simultaneously by $\mathrm{A}_{\bar{1},\bar{2}}$. This rule is valid for discriminants up to $n \leq 5$.
- In all cases the highest degree of coefficients in the discriminant always matches the degree of determinant of real matrix representation of A, in this case 2×2 real matrices (next slide)

Inverses in dimensions $n = p + q \leq 5$

Notes about discriminant

- The discriminant s_m comprises all coefficients a_i of multivector A
- The MV has inverse if discriminant is unequal to zero, $s_m \neq 0$. This imposes the requirement for MV coefficients.
- Because reverse leaves the discriminant invariant, the formula for $n = 2$ can be rewritten as $s_2 = \mathrm{A}\mathrm{A}_{\bar{1},\bar{2}} = \mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}$, from which the left/right inverse formulas follow,

$$\mathrm{A}^{-1} = \mathrm{A}_{\bar{1},\bar{2}}/(\mathrm{A}\mathrm{A}_{\bar{1},\bar{2}}); \qquad \mathrm{A}^{-1} = \mathrm{A}_{\bar{1},\bar{2}}/(\mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}).$$

- The eliminated grades are grade-negated in one of MVs in the self-product. In the example the grades 1 and 2 have been eliminated simultaneously by $\mathrm{A}_{\bar{1},\bar{2}}$. This rule is valid for discriminants up to $n \leq 5$.
- In all cases the highest degree of coefficients in the discriminant always matches the degree of determinant of real matrix representation of A, in this case 2×2 real matrices (next slide)

Inverses in dimensions $n = p + q \leq 5$

Notes about discriminant

- The discriminant s_m comprises all coefficients a_i of multivector A
- The MV has inverse if discriminant is unequal to zero, $s_m \neq 0$. This imposes the requirement for MV coefficients.
- Because reverse leaves the discriminant invariant, the formula for $n = 2$ can be rewritten as $s_2 = \mathrm{A}\mathrm{A}_{\bar{1},\bar{2}} = \mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}$, from which the left/right inverse formulas follow,

$$\mathrm{A}^{-1} = \mathrm{A}_{\bar{1},\bar{2}}/(\mathrm{A}\mathrm{A}_{\bar{1},\bar{2}}); \qquad \mathrm{A}^{-1} = \mathrm{A}_{\bar{1},\bar{2}}/(\mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}).$$

- The eliminated grades are grade-negated in one of MVs in the self-product. In the example the grades 1 and 2 have been eliminated simultaneously by $\mathrm{A}_{\bar{1},\bar{2}}$. This rule is valid for discriminants up to $n \leq 5$.
- In all cases the highest degree of coefficients in the discriminant always matches the degree of determinant of real matrix representation of A, in this case 2×2 real matrices (next slide)

Inverses in dimensions $n = p + q \leq 5$

Period-8 table of irreducible matrix representations of GA for $Cl_{p,q}$, where p and q indicate positive and negative signature.

$p \backslash q$	0	1	2	3	4	5	6	7
0	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	${}^2\mathbb{H}$	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$
1	${}^2\mathbb{R}$	$\mathbb{R}(2)$	$\mathbb{C}(2)$	$\mathbb{H}(2)$	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$
2	$\mathbb{R}(2)$	${}^2\mathbb{R}(2)$	$\mathbb{R}(4)$	$\mathbb{C}(4)$	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$
3	$\mathbb{C}(2)$	$\mathbb{R}(4)$	${}^2\mathbb{R}(4)$	$\mathbb{R}(8)$	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$
4	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$	$\mathbb{R}(16)$	$\mathbb{C}(16)$	$\mathbb{H}(16)$	${}^2\mathbb{H}(16)$
5	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$	${}^2\mathbb{R}(16)$	$\mathbb{R}(32)$	$\mathbb{C}(32)$	$\mathbb{H}(32)$
6	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$	$\mathbb{R}(32)$	${}^2\mathbb{R}(32)$	$\mathbb{R}(64)$	$\mathbb{C}(64)$
7	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$	$\mathbb{C}(32)$	$\mathbb{R}(64)$	${}^2\mathbb{R}(64)$	$\mathbb{R}(128)$

The left superscript denotes direct sum of matrices, ${}^2\mathbb{R}(2) \equiv \mathbb{R}(2) \oplus \mathbb{R}(2)$.

Inverses in dimensions $n = p + q \leq 5$

Three dimensional algebras: $Cl_{2,1}$ case

- Our goal is to eliminate as many grades as possible by forming suitable self-products, for example, for $Cl_{2,1}$ algebra.
- The $\mathrm{A} = a_0 + a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 + a_4\mathbf{e}_{12} + a_5\mathbf{e}_{13} + a_6\mathbf{e}_{23} + a_7\mathbf{e}_{123}$ product with $\mathrm{A}_{\bar{1},\bar{2}}$ eliminates grades 1 and 2. The result is the new multivector $\mathrm{B} = \mathrm{A}\mathrm{A}_{\bar{1},\bar{2}} = b_0 + b_7\mathbf{e}_{123}$ which consists of scalar and grade-3 element. In particular, $b_0 = a_0^2 - a_1^2 - a_2^2 + a_3^2 + a_4^2 - a_5^2 - a_6^2 + a_7^2$ and $b_7 = -2a_3a_4 + 2a_2a_5 - 2a_1a_6 + 2a_0a_7$.
- Repeating the grade negation procedure for grade-3 then yields left-handed discriminant

$$s_4 = \mathrm{B}\mathrm{B}_{\bar{3}} = \mathrm{A}\mathrm{A}_{\bar{1},\bar{2}}(\mathrm{A}\mathrm{A}_{\bar{1},\bar{2}})_{\bar{3}} = b_0^2 - b_7^2.$$

- Reversion of s_4 immediately gives the A-right-handed discriminant

$$s_4' = (\mathrm{A}_{\bar{1},\bar{2}}\mathrm{A})_{\bar{3}}\mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}$$

that contains four MVs (next slide).

Inverses in dimensions $n = p + q \leq 5$

Period-8 table of irreducible matrix representations of GA for $Cl_{p,q}$, where p and q indicate positive and negative signature.

$p \backslash q$	0	1	2	3	4	5	6	7
0	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	${}^2\mathbb{H}$	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$
1	${}^2\mathbb{R}$	$\mathbb{R}(2)$	$\mathbb{C}(2)$	$\mathbb{H}(2)$	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$
2	$\mathbb{R}(2)$	${}^2\mathbb{R}(2)$	$\mathbb{R}(4)$	$\mathbb{C}(4)$	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$
3	$\mathbb{C}(2)$	$\mathbb{R}(4)$	${}^2\mathbb{R}(4)$	$\mathbb{R}(8)$	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$
4	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$	$\mathbb{R}(16)$	$\mathbb{C}(16)$	$\mathbb{H}(16)$	${}^2\mathbb{H}(16)$
5	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$	${}^2\mathbb{R}(16)$	$\mathbb{R}(32)$	$\mathbb{C}(32)$	$\mathbb{H}(32)$
6	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$	$\mathbb{R}(32)$	${}^2\mathbb{R}(32)$	$\mathbb{R}(64)$	$\mathbb{C}(64)$
7	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$	$\mathbb{C}(32)$	$\mathbb{R}(64)$	${}^2\mathbb{R}(64)$	$\mathbb{R}(128)$

The left superscript denotes direct sum of matrices, ${}^2\mathbb{R}(2) \equiv \mathbb{R}(2) \oplus \mathbb{R}(2)$.

Inverses in dimensions $n = p + q \leq 5$

Three dimensional algebras: $Cl_{2,1}$ case (continued)

- For other 3D algebras, namely $Cl_{3,0}$, $Cl_{1,2}$, and $Cl_{0,3}$ the same formula gives different fully expanded expressions, with other signs of individual coefficients a_i.
- The discriminant is determined by squares of two coefficients, b_0^2 and b_7^2, the difference of which should not be zero for invertible multivector A to exist.

Inverses in dimensions $n = p + q \leq 5$

Four dimensional algebras: $n = p + q = 4$

- In this case elimination of grades can proceed in two alternative ways.
- 1) Negate simultaneously grades 1 and 2 and then in the next step the grades 3 and 4, i. e. $s_4 = \mathrm{A}\mathrm{A}_{\bar{1},\bar{2}}(\mathrm{A}\mathrm{A}_{\bar{1},\bar{2}})_{\bar{3},\bar{4}}$
- 2) Eliminate 2 and 3, and then 1 and 4 grades, i. e. $s_4 = \mathrm{A}\mathrm{A}_{\bar{2},\bar{3}}(\mathrm{A}\mathrm{A}_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$.
- Similar elimination can be done in right-hand versions, $s_4' = (\mathrm{A}_{\bar{2},\bar{3}}\mathrm{A})_{\bar{1},\bar{4}}\mathrm{A}_{\bar{2},\bar{3}}\mathrm{A} = (\mathrm{A}_{\bar{1},\bar{2}}\mathrm{A})_{\bar{3},\bar{4}}\,\mathrm{A}_{\bar{1},\bar{2}}\mathrm{A}$.
- Despite different forms, expanded explicit expressions for discriminants s_m and s_m' in fact always appear to be equal, $s_m = s_m'$.
- The appearance of symbol A as much as four times in the products s_4 and s_4' is again what we expect from the matrix representation for $n = 4$ algebras, (next slide) namely, the highest degree of determinant polynomial should be of degree 4 as well, in accord with 4×4 matrix representation $\mathbb{R}(4)$.

Inverses in dimensions $n = p + q \leq 5$

Period-8 table of irreducible matrix representations of GA for $Cl_{p,q}$, where p and q indicate positive and negative signature.

$p \backslash q$	0	1	2	3	4	5	6	7
0	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	${}^2\mathbb{H}$	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$
1	${}^2\mathbb{R}$	$\mathbb{R}(2)$	$\mathbb{C}(2)$	$\mathbb{H}(2)$	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$
2	$\mathbb{R}(2)$	${}^2\mathbb{R}(2)$	$\mathbb{R}(4)$	$\mathbb{C}(4)$	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$
3	$\mathbb{C}(2)$	$\mathbb{R}(4)$	${}^2\mathbb{R}(4)$	$\mathbb{R}(8)$	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$
4	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$	$\mathbb{R}(16)$	$\mathbb{C}(16)$	$\mathbb{H}(16)$	${}^2\mathbb{H}(16)$
5	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$	${}^2\mathbb{R}(16)$	$\mathbb{R}(32)$	$\mathbb{C}(32)$	$\mathbb{H}(32)$
6	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$	$\mathbb{R}(32)$	${}^2\mathbb{R}(32)$	$\mathbb{R}(64)$	$\mathbb{C}(64)$
7	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$	$\mathbb{C}(32)$	$\mathbb{R}(64)$	${}^2\mathbb{R}(64)$	$\mathbb{R}(128)$

The left superscript denotes direct sum of matrices, ${}^2\mathbb{R}(2) \equiv \mathbb{R}(2) \oplus \mathbb{R}(2)$.

Inverses in dimensions $n = p + q \leq 5$

Five dimensional algebras: $n = p + q = 5$

- 1) Eliminate simultaneously grades 2 and 3
- 2) Then eliminate grades 1 and 4
- 3) Finally eliminate grade 5. Apart from the last step this sequence is exactly the same as the second alternative of $n = 4$ case.
- The final result is

$$s'_8 = (\mathrm{FA})_{\bar{5}}\mathrm{F}{\color{red}\mathrm{A}} \quad \text{with} \quad \mathrm{F} = (\mathrm{A}_{\bar{2},\bar{3}}\mathrm{A})_{\bar{1},\bar{4}}\mathrm{A}_{\bar{2},\bar{3}}$$

  for right-hand sided form of the discriminant and

$$s_8 = {\color{red}\mathrm{A}}\mathrm{D}\,(\mathrm{AD})_{\bar{5}} \quad \text{with } \mathrm{D} = \mathrm{A}_{\bar{2},\bar{3}}(\mathrm{A}\mathrm{A}_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$$

  for left-hand discriminant.
- Matrix representation of five dimensional Clifford algebras corresponds to real ${}^2\mathbb{R}(4)$ matrix (next slide) the the determinant of which is a polynomial of degree 8. This is in accord with the structure of discriminant $s_8 = s'_8$ where the A occurs as much as eight times.

Inverses in dimensions $n = p + q \leq 5$

Period-8 table of irreducible matrix representations of GA for $Cl_{p,q}$, where p and q indicate positive and negative signature.

$p \backslash q$	0	1	2	3	4	5	6	7
0	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	${}^2\mathbb{H}$	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$
1	${}^2\mathbb{R}$	$\mathbb{R}(2)$	$\mathbb{C}(2)$	$\mathbb{H}(2)$	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$
2	$\mathbb{R}(2)$	${}^2\mathbb{R}(2)$	$\mathbb{R}(4)$	$\mathbb{C}(4)$	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$
3	$\mathbb{C}(2)$	$\mathbb{R}(4)$	${}^2\mathbb{R}(4)$	$\mathbb{R}(8)$	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$
4	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$	$\mathbb{R}(16)$	$\mathbb{C}(16)$	$\mathbb{H}(16)$	${}^2\mathbb{H}(16)$
5	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$	${}^2\mathbb{R}(16)$	$\mathbb{R}(32)$	$\mathbb{C}(32)$	$\mathbb{H}(32)$
6	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$	$\mathbb{R}(32)$	${}^2\mathbb{R}(32)$	$\mathbb{R}(64)$	$\mathbb{C}(64)$
7	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$	$\mathbb{C}(32)$	$\mathbb{R}(64)$	${}^2\mathbb{R}(64)$	$\mathbb{R}(128)$

The left superscript denotes direct sum of matrices, ${}^2\mathbb{R}(2) \equiv \mathbb{R}(2) \oplus \mathbb{R}(2)$.

Inverses in dimensions $n = p + q \leq 5$

The case of even multivectors

- Even multivectors make subalgebra
- There is an isomorphisms

$$Cl^{+}_{p,q+1} \cong Cl_{p,q}, \quad Cl^{+}_{p+1,q} \cong Cl_{q,p}$$

- This can help to get inverse for MV of lower dimensional algebra once formula of higher dimension algebra is known.
- Example for $p + q = 4$. Formula of inverse of general MV for $n = 4$ is

$$\mathrm{A}^{-1} = \frac{\mathrm{A}_{\bar{2},\bar{3}}(\mathrm{A}\mathrm{A}_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}}{\mathrm{A}\mathrm{A}_{\bar{2},\bar{3}}(\mathrm{A}\mathrm{A}_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}}$$

- If A consists only of even grades then A is invariant to odd grade negations, so that the above formula for even MV's reduces to

$$\mathrm{A}^{-1} = \frac{\mathrm{A}_{\bar{2}}(\mathrm{A}\mathrm{A}_{\bar{2}})_{\bar{4}}}{\mathrm{A}\mathrm{A}_{\bar{2}}(\mathrm{A}\mathrm{A}_{\bar{2}})_{\bar{4}}}$$

Inverses in dimensions $n = p + q \leq 5$

The case of even multivectors (continued)

- Remember that even grade elements of $Cl_{1,3}$ go to vectors and bivectors of $Cl_{3,0}$ and the pseudoscalar I_4 goes to pseudoscalar I_3. Thus, in $Cl_{3,0}$ algebra notation the last formula should be rewritten as

$$\mathsf{A}^{-1} = \frac{\mathsf{A}_{\bar{1},\bar{2}}(\mathsf{A}\mathsf{A}_{\bar{1},\bar{2}})_{\bar{3}}}{\mathsf{A}\mathsf{A}_{\bar{1},\bar{2}}(\mathsf{A}\mathsf{A}_{\bar{1},\bar{2}})_{\bar{3}}}$$

  which coincides with the result for $p + q = 3$
- An alternative formulas for even multivectors exist, which includes even MV multiplication by vector (next slide)
- The inverse of even MV for $p + q = 6$ can be written as a self-negated product

Inverses in dimensions $n = p + q \leq 5$

The case of even multivectors (continued)

$Cl_{p,q}$	$\left(\mathrm{A}^+\right)^{-1}$	
$p+q=2$	$\dfrac{B\left(\mathrm{A}^+B\right)}{\left(\mathrm{A}^+B\right)^2}$,	$B = (\mathbf{e}_i)_{\bar{1}} = -\mathbf{e}_i$
$p+q=3$	$\dfrac{C\left(\mathrm{A}^+C\right)}{\left(\mathrm{A}^+C\right)^2}$,	$C = (\mathrm{A}^+\mathbf{e}_i)_{\bar{1}}$
$p+q=4$	$\dfrac{C\left(\mathrm{A}^+C\right)}{\left(\mathrm{A}^+C\right)^2}$,	$C = (\mathrm{A}^+\mathbf{e}_i)_{\bar{1}}$
$p+q=5$	$\dfrac{D\left(\mathrm{A}^+D\right)}{\left(\mathrm{A}^+D\right)^2}$,	$D = (\mathrm{A}^+\mathbf{e}_i)_{\bar{3}}\left(\mathrm{A}^+(\mathrm{A}^+\mathbf{e}_i)_{\bar{3}}\right)_{\bar{1}}$
$p+q=6$	$\dfrac{D\left(\mathrm{A}^+D\right)_{\bar{6}}}{\left(\mathrm{A}^+D\right)\left(\mathrm{A}^+D\right)_{\bar{6}}}$,	$D = (\mathrm{A}^+\mathbf{e}_i)_{\bar{3}}\left(\mathrm{A}^+(\mathrm{A}^+\mathbf{e}_i)_{\bar{3}}\right)_{\bar{1}}$

Inverses in dimensions $n = p + q \leq 5$

The case of even multivectors (continued)

The denominators in inverse formulas are the discriminants of respective multivectors

$$s_2 = \big(\mathrm{A}^+ B\big)^2 / \mathbf{e}_i^2, \qquad s_4' = \big(\mathrm{A}^+ C\big)^2 / \mathbf{e}_i^2, \qquad s_4 = \big(\mathrm{A}^+ D\big)^2 / \mathbf{e}_i^4$$

and

$$s_8 = \big(\mathrm{A}^+ D\big)\big(\mathrm{A}^+ D\big)_{\bar{6}} / \mathbf{e}_i^4 .$$

Here explicit $\mathbf{e}_i^2 = \pm 1$ and $\mathbf{e}_i^4 = +1$ factors take into account the correct sign and norm once the orthonormal base vector $\mathbf{e}_i$ is replaced by arbitrary unnormalized and nonisotropic vector v.

Can grade elimination procedure be extended beyond $n = 5$?

The short answer is 'yes' if we allow multilinear combinations of grade-negated self-products with properly chosen coefficients.

Inverse in dimension $n = p + q = 6$

Why single self-negated product does not work in $p + q = 6$?

- The culprit appear to be the grade-4 blades. They cannot be eliminated by single sef-product!
- For $n = 6$ we can simultaneously eliminate either grades 1, 2, 5 and 6 (then 0, 3 and 4 grades survive) or, alternatively, the grades 2, 3 and 6 (then grades 0, 1, 4 and 5 persist).
- There exists a subalgebra with base formed by elements $\{1, \mathbf{e}_{1256}, \mathbf{e}_{1346}, \mathbf{e}_{2345}\}$ such that any self product of multivector

$$\mathrm{B} = a_0 + a_{47}\mathbf{e}_{1256} + a_{49}\mathbf{e}_{1346} + a_{52}\mathbf{e}_{2345}$$

  by grade-negated B will yield new MV having at least one grade-4 base element present.
- There is only one such an "uneliminatible" grade part in $p + q = 6$ case.

We will start by trying to eliminate this special grade-4 part.

Inverse in dimension $n = p + q = 6$

How many terms we need in product?

$p \backslash q$	0	1	2	3	4	5	6	7
0	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	${}^2\mathbb{H}$	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$
1	${}^2\mathbb{R}$	$\mathbb{R}(2)$	$\mathbb{C}(2)$	$\mathbb{H}(2)$	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$
2	$\mathbb{R}(2)$	${}^2\mathbb{R}(2)$	$\mathbb{R}(4)$	$\mathbb{C}(4)$	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$
3	$\mathbb{C}(2)$	$\mathbb{R}(4)$	${}^2\mathbb{R}(4)$	$\mathbb{R}(8)$	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$
4	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	${}^2\mathbb{R}(8)$	$\mathbb{R}(16)$	$\mathbb{C}(16)$	$\mathbb{H}(16)$	${}^2\mathbb{H}(16)$
5	${}^2\mathbb{H}(2)$	$\mathbb{H}(4)$	$\mathbb{C}(8)$	$\mathbb{R}(16)$	${}^2\mathbb{R}(16)$	$\mathbb{R}(32)$	$\mathbb{C}(32)$	$\mathbb{H}(32)$
6	$\mathbb{H}(4)$	${}^2\mathbb{H}(4)$	$\mathbb{H}(8)$	$\mathbb{C}(16)$	$\mathbb{R}(32)$	${}^2\mathbb{R}(32)$	$\mathbb{R}(64)$	$\mathbb{C}(64)$
7	$\mathbb{C}(8)$	$\mathbb{H}(8)$	${}^2\mathbb{H}(8)$	$\mathbb{H}(16)$	$\mathbb{C}(32)$	$\mathbb{R}(64)$	${}^2\mathbb{R}(64)$	$\mathbb{R}(128)$

We need 8 term product. Too many. Fortunately the determinant of matrix representation of scalar+grade-4 multivector is a square!, i. e. $\det(\langle A \rangle_4) = (\cdots)^2$. Only 4 term product multilinear combination should eliminate grade-4 part.

Inverse in dimension $n = p + q = 6$

Pattern of multilinear combination

- Form prototype of a geometric product of the form $s_{4+4} = s_{4f} + s_{4g} =$

$$b_1 B f_5\Big(f_4(B) f_3\big(f_2(B) f_1(B)\big)\Big) + b_2 B g_5\Big(g_4(B) g_3\big(g_2(B) g_1(B)\big)\Big),$$

  where f_j, g_j denotes yet undetermined grade-4 negations and b_1, b_2 are unknown scalar coefficients of the multilinear combination.
- Not unique. The other three possible choices, however, yield the same answer.

Calculate explicit symbolic form of square root of the determinant of matrix representation of the multivector $\mathrm{B} = a_0 + a_{47}\mathbf{e}_{1256} + a_{49}\mathbf{e}_{1346} + a_{52}\mathbf{e}_{2345}$ and compare it with all possible negations of grade-4 substituted for f_j and g_j.

Inverse in dimension $n = p + q = 6$

Modelling the grade negation

- Multiply the grade-4 part by unknown coefficients p_{4jk}, where the index j denotes the involution number f_j, g_j in the self-product and k is the term in multilinear combination (in $p + q = 6$ case $k = 1$ for f and $k = 2$ for g).
- The coefficients p_{ijk} acquire the values ± 1 only, where -1 means that involution that changes sign of grade i is applied and $+1$ means that grade-i remains unchanged.
- The answer: $\{b_1 = -2/3,\ b_2 = -1/3,\ p_{411} = -1,\ p_{412} = 1,\ p_{421} = -1,\ p_{422} = 1,\ p_{431} = -1,\ p_{432} = -1,\ p_{441} = -1,\ p_{442} = 1,\ p_{451} = -1,\ p_{452} = 1\}$ and $\{b_1 = -1/3,\ b_2 = -2/3,\ p_{411} = 1,\ p_{412} = -1,\ p_{421} = 1,\ p_{422} = -1,\ p_{431} = -1,\ p_{432} = -1,\ p_{441} = 1,\ p_{442} = -1,\ p_{451} = 1,\ p_{452} = -1\}$.
- Check the answer with the most general scalar + grade-4 multivector.

Inverse in dimension $n = p + q = 6$

The full discriminant

- Repeat the procedure for involutions that consecutively eliminate all other grades. How to double the number of multivectors?
- Simultaneusly either grades 1, 2, 5 and 6 (then grades 0, 3 and 4 remain), or, alternatively grades 2, 3 and 6 (then grades 0, 1, 4 and 5 remain) can be simultaneously eliminated. All in all there are $\frac{6!}{2!4!} + \frac{6!}{3!3!} + 1 = 36$ elements of grades 2, 3 and 6 which can be eliminated simultaneously. This is more than 28 total elements of grades 1, 2, 5 and 6 that need to be eliminated.
- Therefore, if we replace B in s_{4+4} pattern by self-negated product $\mathrm{A}\mathrm{A}_{\bar{2},\bar{3},\bar{6}}$ we will have eight required terms.
- Then we find the other involutions using the multivector $a_0 + a_1\mathbf{e}_1 + a_{22}\mathbf{e}_{123} + a_{47}\mathbf{e}_{1256}$ and check result with general MV.

More that 300 valid solutions. Among them there is just one solution with minimal number of negated MVs equal to 10.

Inverse in dimension $n = p + q = 6$

The answer

$Cl_{p,q}$	A^{-1}	
$p+q=0$	$\frac{B}{AB}$,	$B = 1$
$p+q=1$	$\frac{B(AB)_{\bar{1}}}{AB\,(AB)_{\bar{1}}} = \frac{\widehat{A}}{A\widehat{A}}$,	$B = 1$
$p+q=2$	$\frac{C}{AC} = \frac{\widehat{\widetilde{A}}}{A\widehat{\widetilde{A}}}$,	$C = A_{\bar{1},\bar{2}}$
$p+q=3$	$\frac{C(AC)_{\bar{3}}}{AC\,(AC)_{\bar{3}}} = \frac{\widehat{\widetilde{A}}\widehat{A}\widetilde{A}}{A\widehat{\widetilde{A}}\widehat{A}\widetilde{A}}$,	$C = A_{\bar{1},\bar{2}}$
$p+q=4$	$\frac{D}{AD}$,	$D = A_{\bar{2},\bar{3}}(AA_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$ or $D = A_{\bar{1},\bar{2}}(AA_{\bar{1},\bar{2}})_{\bar{3},\bar{4}}$
$p+q=5$	$\frac{D(AD)_{\bar{5}}}{AD\,(AD)_{\bar{5}}}$,	$D = A_{\bar{2},\bar{3}}(AA_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$
$p+q=6$	$\frac{G}{AG}$,	$G = \frac{1}{3}A_{\bar{2},\bar{3},\bar{6}}\Big(H(HH)_{\bar{1},\bar{4},\bar{5}} + 2\Big(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}\Big)_{\bar{4}}\Big)$

or $\quad G = \frac{1}{3}A_{\bar{2},\bar{3},\bar{6}}\Big(\Big(H(HH_{\bar{1},\bar{5}})_{\bar{4}}\Big)_{\bar{1},\bar{5}} + 2\Big(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}}\Big)_{\bar{1},\bar{4}}\Big)$

with $\quad H = AA_{\bar{2},\bar{3},\bar{6}} = A\widetilde{A}$

Conclusions

- Conjecture. The inverse of a general MV of arbitrary Clifford algebra can be always expressed as a multilinear combination of proper number of specially constructed self-negated products, where in the discriminant expression the initial MV stays in the left-most or right-most position.
- The number of the multivectors in the negated products is determined by dimension of matrix representation of the algebra.
- Hypothesis. The minimal number of terms in the multilinear combination is related to the number of "uneliminatable" grades in the self product. There are no such grades in algebras with $p + q \leq 3$. There is only one grade, namely the grade-4, in algebras of dimension 4, 5, 6 and 7. The dimensions 4 and 5 are exceptional (not enough indices to form cyclic subgroup of the form scalar + grade-4). The algebras with $p + q = 8$ and $p + q = 9$ have two such special grades, 4 and 8.
- Computational performance? Multiplication of 8 general multivectors requires $(2^6)^8 = 281\,474\,976\,710\,656$ geometric multiplications of base elements in 6-dimensional algebra.

Conclusions

- Conjecture. The inverse of a general MV of arbitrary Clifford algebra can be always expressed as a multilinear combination of proper number of specially constructed self-negated products, where in the discriminant expression the initial MV stays in the left-most or right-most position.
- The number of the multivectors in the negated products is determined by dimension of matrix representation of the algebra.
- Hypothesis. The minimal number of terms in the multilinear combination is related to the number of "uneliminatable" grades in the self product. There are no such grades in algebras with $p + q \leq 3$. There is only one grade, namely the grade-4, in algebras of dimension 4, 5, 6 and 7. The dimensions 4 and 5 are exceptional (not enough indices to form cyclic subgroup of the form scalar + grade-4). The algebras with $p + q = 8$ and $p + q = 9$ have two such special grades, 4 and 8.
- Computational performance? Multiplication of 8 general multivectors requires $(2^6)^8 = 281\,474\,976\,710\,656$ geometric multiplications of base elements in 6-dimensional algebra.

Conclusions

- Conjecture. The inverse of a general MV of arbitrary Clifford algebra can be always expressed as a multilinear combination of proper number of specially constructed self-negated products, where in the discriminant expression the initial MV stays in the left-most or right-most position.
- The number of the multivectors in the negated products is determined by dimension of matrix representation of the algebra.
- Hypothesis. The minimal number of terms in the multilinear combination is related to the number of "uneliminatable" grades in the self product. There are no such grades in algebras with $p + q \leq 3$. There is only one grade, namely the grade-4, in algebras of dimension 4, 5, 6 and 7. The dimensions 4 and 5 are exceptional (not enough indices to form cyclic subgroup of the form scalar + grade-4). The algebras with $p + q = 8$ and $p + q = 9$ have two such special grades, 4 and 8.
- Computational performance? Multiplication of 8 general multivectors requires $(2^6)^8 = 281\,474\,976\,710\,656$ geometric multiplications of base elements in 6-dimensional algebra.

Conclusions

- Conjecture. The inverse of a general MV of arbitrary Clifford algebra can be always expressed as a multilinear combination of proper number of specially constructed self-negated products, where in the discriminant expression the initial MV stays in the left-most or right-most position.
- The number of the multivectors in the negated products is determined by dimension of matrix representation of the algebra.
- Hypothesis. The minimal number of terms in the multilinear combination is related to the number of "uneliminatable" grades in the self product. There are no such grades in algebras with $p + q \leq 3$. There is only one grade, namely the grade-4, in algebras of dimension 4, 5, 6 and 7. The dimensions 4 and 5 are exceptional (not enough indices to form cyclic subgroup of the form scalar + grade-4). The algebras with $p + q = 8$ and $p + q = 9$ have two such special grades, 4 and 8.
- Computational performance? Multiplication of 8 general multivectors requires $(2^6)^8 = 281\,474\,976\,710\,656$ geometric multiplications of base elements in 6-dimensional algebra.

Bibliography

P. Dadbeh,
Inverse and determinant in 0 to 5 dimensional Clifford algebra,
arXiv: 1104.0067 (Mar. 2011).

A. Acus and A. Dargys,
Geometric algebra Mathematica package, 2017,
https://github.com/ArturasAcus/GeometricAlgebra.

A. Acus and A. Dargys,
submitted to Advances in Applied Clifford algebras

Thank You